



Day-2 ML

Self-Correcting Travel Booking System



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Problem Statement



Traditional travel booking is:

- Manual and time-consuming
- Doesn't adapt when constraints are violated
- No intelligent fallback options

Solution: Autonomous agents that self-correct and optimize

Architecture Overview

User Request → Supervisor → Planner → Judge



(self-correction loop)



Human Approval → Book

3 Specialized Agents:

1. Supervisor - Orchestrates workflow
2. Planner - Generates itineraries using tools
3. Judge - Validates & provides feedback

Reflection Pattern

Judge acts as a critic:

-  Validates budget compliance
-  Checks transport availability
-  Verifies hotel booking
-  Rejects & provides specific feedback
-  Triggers revision (max 3 attempts)

Result: System self-corrects without human intervention

Multi-Modal Intelligence



Planner automatically chooses best option:

- Flights - Fast but expensive
- Trains - Budget-friendly alternative
- Car Rentals - Flexible for road trips
- Hotels - Multiple categories (5-star to budget)

Smart pivoting: If flight exceeds budget → automatically tries train

Real-World Example

Scenario: NYC → Boston, \$300 budget

1. Attempt 1: Planner books flight (\$450)
2. Judge: "Budget exceeded by \$150"
3. Attempt 2: Planner pivots to train (\$180)
4. Judge: "Approved - under budget"
5. Human: Confirms booking

Total iterations: 2 (automatic self-correction)

Tech Stack



Framework: LangGraph (state machine for agents) LLM: Ollama (local inference, no API costs) APIs: RapidAPI Booking.com

15 Key Features:

- Typed state management
- Tool calling (flights, trains, hotels, cars)
- Checkpointing for human approval
- Retry logic with exponential backoff

What I Learned

- Reflection pattern enables self-correction
- Multi-agent coordination handles complex workflows
- Tool abstraction makes system extensible
- Human oversight ensures final control
- Production-ready with real API integrations